# ecFactory: Computational Prediction of Metabolic Engineering Targets in Yeast

*Integrating the FSEOF algorithm with ecModels to Predict Viable Engineering Targets*

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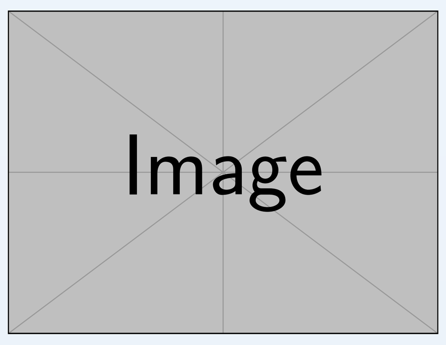
VU Introduction to metabolic modelling (2026S)





## Abstract

Engineering microbial cell factories (MCFs) for the production of valuable chemicals is time- and resource-intensive, typically requiring several years of research and substantial amounts of material to advance a proof-of-concept strain to commercial scale. Genome-scale metabolic models (GEMs) combined with computational algorithms can identify non-intuitive genetic targets, yet existing methods return extensive candidate lists without integrated filtering, and standard GEMs overpredict metabolic capabilities by neglecting enzyme capacity constraints. **ecFactory** is a three-step computational pipeline that leverages *enzyme-constrained metabolic models* (ecModels) to predict a minimal, optimal combination of gene targets for enhanced chemical bioproduction. The application of ecFactory to **103 chemicals** in *Saccharomyces cerevisiae*, reduced ∼85 initial gene candidates per product by **73%** to ∼21 optimal targets. Predictions were experimentally validated for **22 chemicals** across independent studies. Clustering of gene-target profiles revealed **8 product groups** sharing common engineering strategies, building the foundation for rational *platform strain* design.


## Motivation & Problem

**Why is engineering cell factories hard?**

- Metabolism is a **highly interconnected network** modifying selected genes can have non-intuitive cascading effects
- Full MCF development is time and resource intensive
- previous methods of computational methods have limited applicability

**Limitations of existing methods**

- existing methods return extensive lists of target gene candidates, ranked lists still require *arbitrary* user-defined cutoffs
- **Standard GEMs:** overpredict metabolic capabilities because they lack kinetic and regulatory information
- **Kinetic models** (e.g.FSEOF, OptKnock, OptForce) lack integration of additional analysis to reduce number of candidate genes

## The ecFactory Pipeline

ecFactory integrates the **FSEOF algorithm** with **enzyme-constrained models** ecYeastGEM v8.3.4, generated by the GECKO 2.0 Toolbox [2] in three sequential steps:

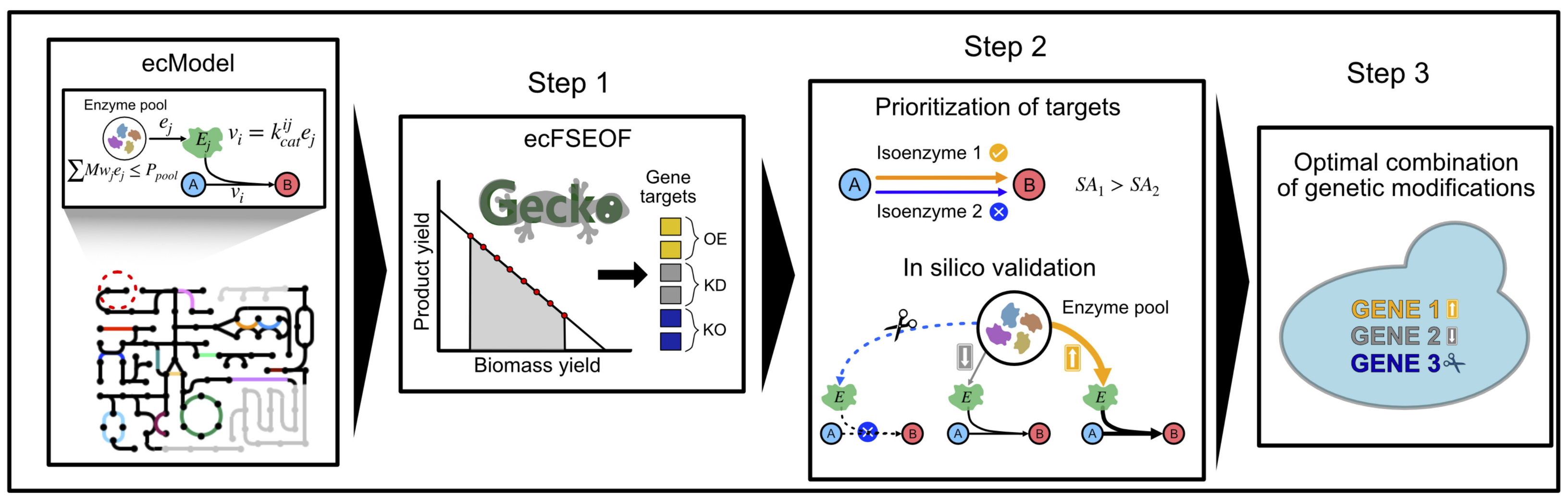



*Fig.1, from [1, Fig.2]: The three-step ecFactory pipeline with the ecModel as Input.*

1. **ecFSEOF** Flux Scanning with Enforced Objective Function on ecModels scores each gene by the direction (OE/KD/KO) and intensity of the required expression change as product secretion is progressively enforced.
2. **Priorization** Discards targets encoding redundant isoenzymes or enzymes with low catalytic efficiency (low specific activity $k_{\mathrm{cat}}$).
3. **Optimal Combination** identifies the *smallest* combination of modifications sufficient to drive the cell from maximal biomass formation to an optimal production regime.

## References


[1] Iván Domenzain et al. "Computational biology predicts metabolic engineering targets for increased production of 103 valuable chemicals in yeast". en. In: *Proc. Natl. Acad. Sci. U. S. A.* 122.9 (Mar. 2025), e2417322122.

[2] Iván Domenzain et al. "Reconstruction of a catalogue of genome-scale metabolic models with enzymatic constraints using GECKO 2.0". en. In: *Nat. Commun.* 13.1 (June 2022), p. 3766.


## Results: Predicting Targets for 103 Chemicals

**Scope:** 103 chemicals across 10 families (amino acids, terpenes, organic acids, aromatics, fatty acids, alcohols, alkaloids, flavonoids, bioamines, stilbenoids) in *S. cerevisiae*.

**Key finding: Enzyme constraints enable reduction of targets**

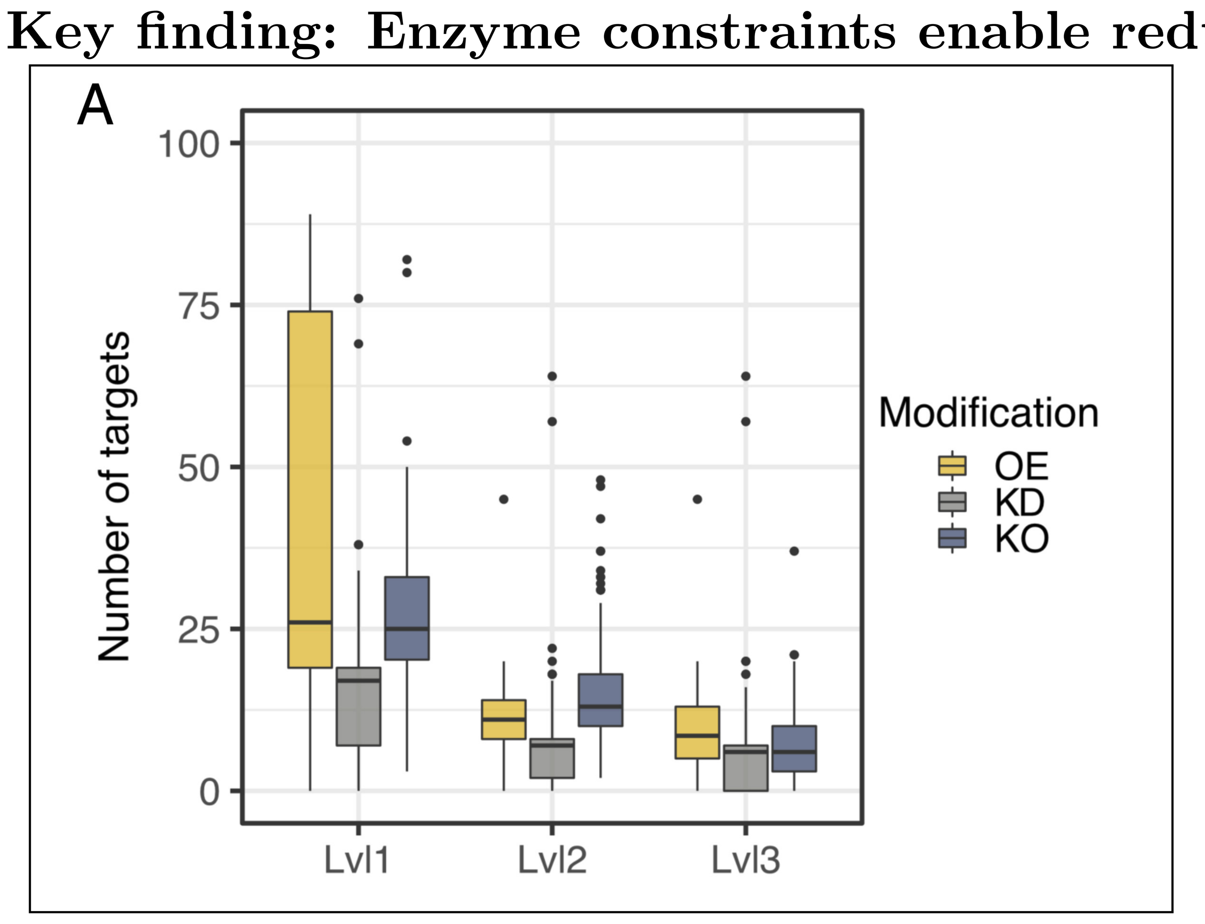



*Fig.2: excerpt from [1, Fig.3A]: Overview of the distribution of predicted number of gene targets per pipeline step. (Levels correspond to the Steps in Fig.1.) OE: overexpression, KD: knockdown, KO: knockout;*

The pipeline reduced the number gene targets from ∼ 85 →∼ 21 representing a **73% reduction**.

**Protein-limited vs. substrate-limited products**

- heterologous products are more often protein-constrained than native products
- OE targets centre on the pentose-phosphate pathway; KD/KO targets on the TCA cycle and oxidative phosphorylation

**Literature Validation**

Predictions matched independently reported engineering targets for **22 chemicals** in published literature and **30 distinct gene targets** were confirmed experimentally across 18 products from different chemical classes

## Towards Platform Strains

**Products with similar engineering needs cluster together**

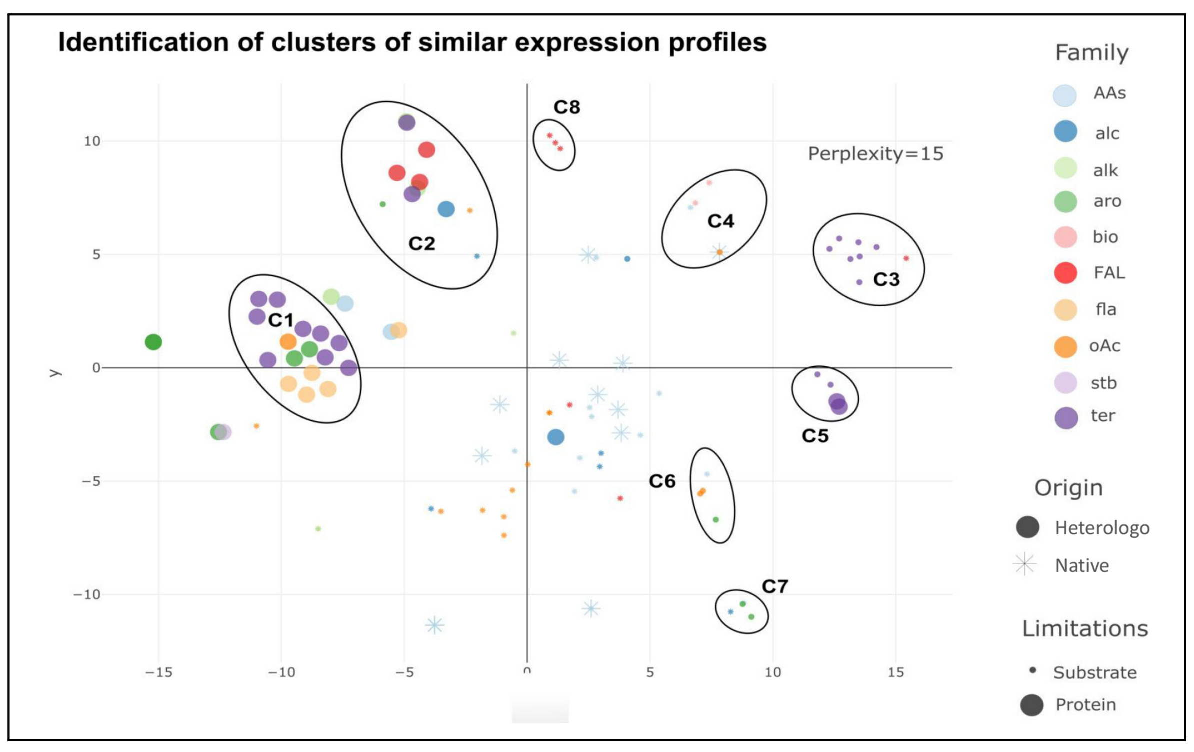



*Fig.3: excerpt from [1, Fig.4B]: t-SNE visualisation of gene-target profiles for 103 chemicals. Eight clusters share common core engineering strategies*

- Engineering strategies are shared beyond chemical families
- Protein burden and metabolic precursor demand are decisive
- The Results highlight the potential for **Platform strains**.